Experiment 2

Measurement of Natural Frequency and Damping of The Single Degree of Freedom Vibratory System

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1. Experiment Purposes

a. Acquire proficiency in multiple prevalent techniques for quantifying the natural frequency and damping characteristics of single degree of freedom vibratory systems.
b. Attain knowledge regarding the appropriate utilization of frequently employed vibration measurement instruments.

2. Experiment Contents

a. Utilize the formula for natural frequency in the context of a single degree of freedom system and estimate for the equivalent mass of the horizontally vibrating table.
b. Document the waveform of free damping vibrations exhibited by the horizontally vibrating table.
c. Conduct measurements to ascertain the magnitude-frequency characteristic of the horizontally vibrating table subjected to simple harmonic excitation.
d. Conduct measurements to determine the phase-frequency characteristic of the horizontally vibrating table under the influence of simple harmonic excitation.
e. Based on the data acquired through measurements, calculate the natural frequency and damping properties of the horizontally vibrating table.

3. Experiment Principles

The single degree of freedom vibration system is a common model, represented by a horizontal vibration table with a table-board, supporting spring piece, and electromagnetic damper (Figure 4). It vibrates horizontally under temporary or continuous disturbance forces. Unlike typical mass-spring systems, this setup lacks known equivalent mass and stiffness, making theoretical natural frequency calculations challenging. Assuming the equivalent mass of the horizontal vibrating table is m_{eq}, a known load added to the system m_0, the equivalent mass can then be calculated through this formula: $\frac{m_{eq} + m_0}{m_{eq}} = \left(\frac{f'}{f''}\right)^2$ where f' is the initial free vibration frequency and f'' is the new frequency after the addition of load.

Damped Vibration:
Initiate vibration using an electric pulse, observing its decay due to damping. Utilize a vibration pickup to convert mechanical signals to voltage, facilitating analysis.

Key Insight: Higher damping leads to faster vibration decay.

$$T_d = \frac{\Delta t}{i}, \quad f_d = \frac{1}{T_d}$$

$$\delta = \frac{1}{i}\ln\left(\frac{X_1}{X_{i+1}}\right) = nT_d, \qquad \zeta = \frac{\delta}{\sqrt{4\pi^2+\delta^2}} \approx \frac{\delta}{2\pi}$$

Amplitude-Frequency Characteristic:
Apply a harmonic force to the system, measuring the amplitude response at various frequencies. Derive an amplitude frequency characteristic curve.
Key Insight: The curve provides insights into the natural frequency and damping ratio.

$$\zeta \approx \frac{\Delta f}{f_n \Delta\varphi_x}$$

phase change $\Delta\varphi_x$(rad)

frequency range△f

Phase-Frequency Characteristic:
Measure phase differences between excitation force and system response at different frequencies. Use this to create a phase difference frequency characteristic curve.
Key Insight: The curve helps calculate the system's damping ratio and natural frequency.

4. Experimental Facilities

Number	Name	The main technical parameters
1	Facility	Natural frequency：about 10Hz Damping ratio：0.01~0.20 variable
2	Relative speed vibration pickup	Working frequency: 2-500Hz displacement: 3mm peak-to-peak value
3	Electromagnetic vibration exciter	Maximum excitation force: 2N Frequency :2~1000Hz
4	Damping excitation power and signal follower	Maximum output current: 1.5A
5	Signal generator and power amplifier	Frequency: 1Hz-6MHz Frequency resolution: 0.01Hz Maximum output current: 1A Maximum power: 15W
6	Microcomputer	A/D、D/A plug-in card is inside

5. Experiment Procedures

i) Initialization and Software Setup:
Begin by launching the "single degree of freedom system" software on the microcomputer to set the stage for data acquisition.

ii) Power Setup:
Power on both the damper excitation and the power amplifier. Adjust the magnet

current to its initial value of 0A.

iii) Equivalent Mass Estimation:
Utilize the "free decay record" function within the software. Gently tap the horizontal vibration table in the horizontal direction. This triggers the display of a free decay curve on the microcomputer screen. Read and record relevant data, including the natural frequency of vibration. Perform this both with and without a 200g weight on the horizontal vibration table to estimate the equivalent mass of the table.

iv) Free Vibration Attenuation Measurement:
Remove the weight from the horizontal vibration table and set the excitation current to 0.6A. Use the "free decay record" function to test the free vibration attenuation. Record the peak values of the attenuation vibration, their corresponding times, and the periodicity i. This data is vital for calculating the damping of the system.

v) Magnitude and Phase Frequency Characteristic Measurement:
Maintain a constant excitation current and select the "simple harmonic excitation vibration" icon. Set up a testing regime where frequencies are tested at 1Hz intervals from 2Hz to 15Hz, and then at 5Hz intervals from 15Hz to 30Hz. Record relevant data, including amplitude ratios and phase differences at various frequencies.

vi) Variation of Damper Excitation Current:
Adjust the damper excitation current to different values (1.0A, 0.8A) and repeat the amplitude and phase frequency characteristic measurements. This allows for a comparison of results under different damper excitation conditions, providing valuable insights into the system's behavior.

2) Formula: $\frac{M_{eq} + M_0}{M_{eq}} = \left(\frac{f'}{f''}\right)^2$ $\qquad \frac{M_{eq} + 0.1}{M_{eq}} = \left(\frac{11.396}{10.840}\right)^2$

$M_{eq} + 0.1 = 1.1052\, M_{eq}$

$0.1052\, M_{eq} = 0.1$

i) Equivalent mass of the horizontal vibration table, $M_{eq} = \underline{0.9506\text{kg}}$

0.6A

ii) System's frequency, f_d

$T_d = \frac{\Delta t}{i} = \frac{0.578 - 0.234}{4} = 0.086$ $\qquad f_d = \frac{1}{T_d} = \frac{1}{0.086} = \underline{11.6279\,Hz}$

iii) Damping ratio, $\xi = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}}$ $\qquad \delta = \frac{1}{i}\ln\left(\frac{x_i}{x_{i+1}}\right) = nT_d$

$\delta = \frac{1}{4}\ln\left(\frac{0.3923}{0.0857}\right) = 0.3803$

$\xi = \frac{0.3803}{\sqrt{4\pi^2 + 0.3803^2}} = \underline{0.0604}$

1.0A

ii) $T_d = \frac{0.496 - 0.152}{4} = 0.086$ $\qquad f_d = \underline{11.6279\,Hz}$

iii) $\delta = \frac{1}{4}\ln\left(\frac{0.5413}{0.0678}\right) = 0.5194$ $\qquad \xi = \underline{0.0824}$

0.8A

ii) $T_d = \frac{0.515 - 0.171}{4} = 0.086$ $\qquad f_d = \underline{11.6279\,Hz}$

iii) $\delta = \frac{1}{4}\ln\left(\frac{0.4358}{0.078}\right) = 0.4301$ $\qquad \xi = \underline{0.0683}$

3) Amplitude-Frequency Characteristics

The greater the amperage, the lower the biggest B_m, the greater the damping ratio hence greater the damping effect.

0.6A

$B_m = 2.111$ $f_m = 11.23\,Hz$ $B_0 = 0.056$ $f_0 = 3\,Hz$

$f_n \approx f_m = 11.23\,Hz$

$$\xi \approx \frac{f_2 - f_1}{2f_m} = \frac{11.81 - 10.72}{2(11.23)} = 0.0488$$

1.0A

$B_m = 1.263$ $f_m = 11.36\,Hz$ $B_0 = 0.056$ $f_1 = 3\,Hz$

$f_n \approx f_m = 11.36\,Hz$

$$\xi = \frac{f_2 - f_1}{2f_m} = \frac{12.36 - 10.51}{2(11.36)} = 0.0814$$

0.8A

$B_m = 1.559$ $f_m = 11.26\,Hz$ $B_0 = 0.056$ $f_0 = 3\,Hz$

$f_n \approx f_m = 11.26\,Hz$

$$\xi = \frac{f_2 - f_1}{2f_m} = \frac{12.17 - 10.64}{2(11.26)} = 0.0679$$

4) Phase-Frequency Characteristics

The greater the amperage, the higher the excitation frequency, f_m.

0.6A

f near the phase resonance point $= f_n$

$f_n = 11.23\,Hz$

$$\xi = \frac{\Delta f}{f_n \Delta \varphi_x} = \frac{11.25 - 11.19}{11.23\left(\frac{5.2}{180}\pi\right)} = 0.0589$$

1.0A

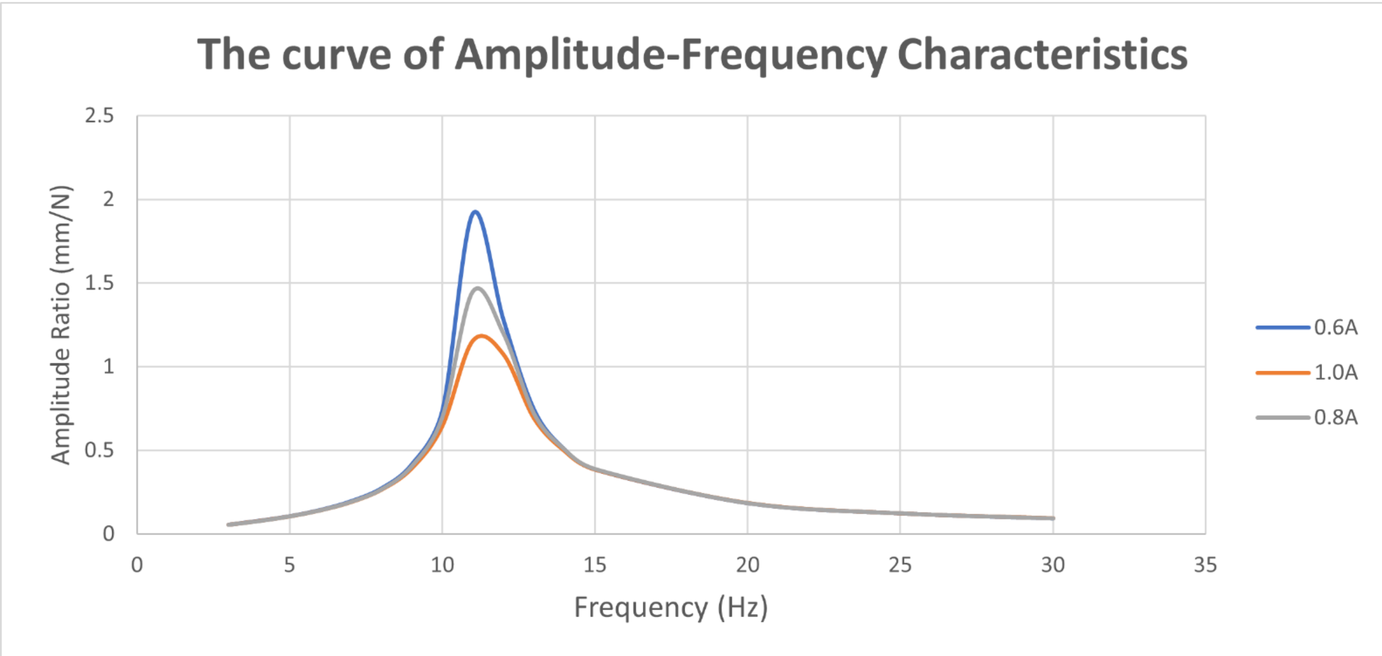
$f_n = 11.35\,Hz$

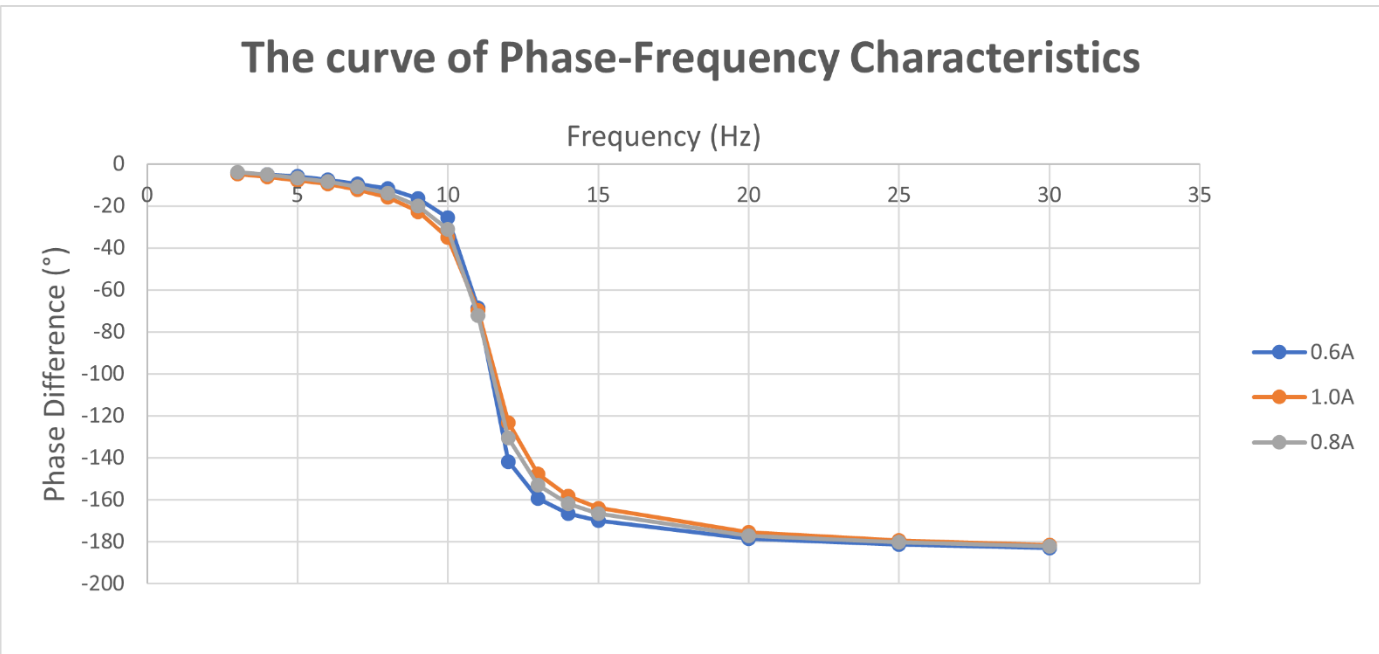
$$\xi = \frac{\Delta f}{f_n \Delta \varphi_x} = \frac{11.38 - 11.30}{11.35\left(\frac{5.4}{180}\pi\right)} = 0.0748$$

$\underline{0.8A}$

$f_n = 11.25\ Hz$

$\xi = \frac{\Delta f}{f_n \psi_x} = \frac{11.30 - 11.20}{11.25\left(\frac{5.7}{180}\pi\right)} = \underline{0.0894}$





5) i) Damped vibration method is the most sensitive to damping effects and Phase-frequency method is the least sensitive to damping effects as damping information is inferred indirectly from phase shifts. That explains why the damping ratio obtained from formula

$$\xi = \frac{\Delta f}{f_n \Delta \varphi_x} \qquad \Delta \varphi_x = \text{phase change (rad)}$$

is not relatively close to the ratios obtained by other methods.

ii) As for applicability, amplitude-frequency method is applicable for harmonic excitations and resonant systems. Phase-frequency method is suitable for non-harmonic excitations and systems with varying phase responses. Lastly, damped vibration method is applicable for observing the system's natural response to initial excitation and decay.

iii) Damping ratios obtained by damped vibration and amplitude-frequency method incenses with current, indicating a stronger damping effect at higher amperages.

单自由度系统固有频率和阻尼的测量

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台面等效质量估算（励磁电流 0A）	增加质量= 100 g	增加质量前/后固有频率=11.396 Hz/10.840 Hz

	励磁电流= 0.6 A		励磁电流= 1.0 A		励磁电流= 0.8 A	
自由衰减振动	X_1	0.3923	X_1	0.5413	X_1	0.4358
	T_1	0.234	T_1	0.152	T_1	0.171
	X_{i+1}	0.0857	X_{i+1}	0.0678	X_{i+1}	0.0380
	T_{i+1}	0.578	T_{i+1}	0.496	T_{i+1}	0.575
	i	4	i	4	i	4
简谐激励振动频率(Hz)	振幅比(mm/N)	相位差(°)	振幅比(mm/N)	相位差(°)	振幅比(mm/N)	相位差(°)
3	0.056	-4.1	0.056	-4.9	0.056	-4.0
4	0.080	-5.2	0.079	-6.1	0.078	-5.0
5	0.1070	-5.8	0.106	-7.8	0.106	-6.8
6	0.1440	-7.5	0.142	-9.6	0.142	-8.5
7	0.1960	-9.4	0.191	-12.3	0.192	-10.8
8	0.2730	-11.8	0.265	-16.0	0.267	-14.0
9	0.4170	-16.4	0.394	-22.7	0.406	-20.0
10	0.7390	-25.7	0.647	-35.1	0.699	-31.1
11	1.9160	-68.6	1.157	-69.7	1.480	-72.2
12	1.281	-141.8	1.075	-123.2	1.199	-130.6
13	0.7484	-159.4	0.692	-147.8	0.720	-153.0
14	0.506	-166.6	0.497	-158.2	0.506	-162.0
15	0.386	-170.0	0.386	-164.0	0.389	-166.7
20	0.185	-178.6	0.186	-175.5	0.185	-177.1
25	0.124	-181.3	0.124	-179.3	0.124	-180.3
30	0.094	-183.0	0.095	-181.6	0.094	-182.2
相频曲线	频率	相位差	频率	相位差	频率	相位差
(f_n, 约-90°)	11.23 Hz	-90.2	11.35 Hz	-90.0	11.25 Hz	-90.1
(f, Φ_1)	11.19 Hz	-87.3	11.30 Hz	-87.0	11.20 Hz	-87.5
(f, Φ_2)	11.25 Hz	-92.5	11.38 Hz	-92.4	11.30 Hz	-93.2
幅频曲线	频率	振幅比	频率	振幅比	频率	振幅比
(f_m, B_m)	11.23 Hz ~~90.5~~	2.111	11.36 Hz	1.263	11.26 Hz	1.559
(f_1, 0.707B_m)	10.72 Hz	1.489	10.51 Hz	0.885	10.64 Hz	1.098
(f_2, 0.707B_m)	11.81 Hz	1.490	12.36 Hz	0.910	12.17 Hz	1.097

Φ_1的取值约在-87°， Φ_2的取值约在-93°